GAD-MambaUNet: Direction-Group Mamba with Gradient-Adaptive DINOv3 Distillation for Lightweight Medical Image Segmentation

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Abstract*—Lightweight medical image segmentation requires accurate delineation of subtle lesion boundaries under strict computational constraints. Convolutional encoder–decoder networks are effective in preserving local details, but their limited receptive fields make it difficult to capture long-range dependencies. Transformer-based methods improve global context modeling, yet often introduce considerable computational and memory overhead. Visual state-space models, such as Mamba, provide an efficient alternative for global modeling with linear complexity, and grouped Mamba further reduces computation by partitioning feature channels into multiple groups. However, existing grouped multi-directional scanning strategies usually process different channel groups and scanning directions independently before fusion, which limits the exchange of complementary contextual information. To address this limitation, we propose GAD-MambaUNet, an asymmetric local–global segmentation network for lightweight medical image segmentation. The proposed network uses lightweight convolutional blocks in shallow stages to preserve fine boundary details and introduces Direction-Group Graph Selective Scan (DG-GSS) blocks in deeper stages for efficient contextual modeling. In each DG-GSS block, direction–group feature responses are represented as graph nodes, and structured message passing is performed before multi-directional fusion, enabling interaction across scanning directions and feature subspaces. To further enhance semantic representation without increasing inference cost, a frozen pretrained DINOv3 model is used only during training as a semantic teacher, and its semantic priors are transferred to the compact student through Gradient-Adaptive Distillation. Experiments on multiple public medical image segmentation datasets demonstrate that GAD-MambaUNet achieves an average Dice score of *XX.XX*

***Index Terms*—medical image segmentation, lightweight networks, state-space models, DINOv3, knowledge distillation**

I. Introduction

Accurate medical image segmentation is fundamental to computer-aided diagnosis, treatment planning, and quantitative disease assessment. Given a medical image, the goal is to delineate clinically relevant structures at the pixel level while preserving their boundaries and spatial extent. This task is challenging because medical targets are often small, irregular, weakly contrasted, or visually similar to surrounding tissue. A useful segmentation model must therefore combine fine-grained local evidence, which is essential for boundary recovery, with sufficient contextual information to distinguish ambiguous foreground and background regions.

U-Net [1] established the dominant encoder–decoder paradigm for medical image segmentation. Its contracting path extracts increasingly abstract semantic representations, while skip connections transfer high-resolution encoder features to the decoder for spatially precise reconstruction. Building on this design, numerous CNN-based methods have improved feature fusion, attention modeling, boundary refinement, and multi-scale decoding. For example, PraNet [2] uses parallel partial decoding and reverse attention for polyp segmentation, whereas UACANet [3] introduces uncertainty-aware context attention. These methods demonstrate the importance of combining semantic understanding with accurate localization.

Despite their success, convolutional networks have an intrinsic limitation: their local receptive fields do not directly capture long-range dependencies. This limitation becomes evident when boundary interpretation requires global lesion shape, long-range boundary continuity, or contextual contrast between foreground targets and surrounding tissues. Transformer-based networks, such as TransUNet [4], address this issue by introducing self-attention for global feature interaction. However, global attention and large pretrained backbones can substantially increase parameter count, memory usage, and computational cost. Thus, stronger contextual modeling does not automatically translate into a practical solution when inference resources are limited.

As medical imaging solutions increasingly move from laboratory-centered analysis to point-of-care applications, segmentation accuracy is no longer the only design objective. Parameter count, computational complexity, memory footprint, and inference efficiency also determine whether a model can be deployed in realistic clinical or edge-device settings. Therefore, an effective lightweight segmentation model should not only improve segmentation accuracy, but also maintain a favorable balance between accuracy and deployment cost.

This transition has motivated lightweight medical segmentation networks based on depth-wise convolution, compact channel schedules, grouped operators, and efficient token mixing. UNeXt [5], CMUNeXt [6], EGE-UNet [7], and EMCAD [8] are representative examples. In particular, MK-UNet [9] uses multi-kernel depth-wise convolution and lightweight decoder attention to retain local multi-scale modeling at a compact cost. Nevertheless, lightweight CNN-dominated backbones

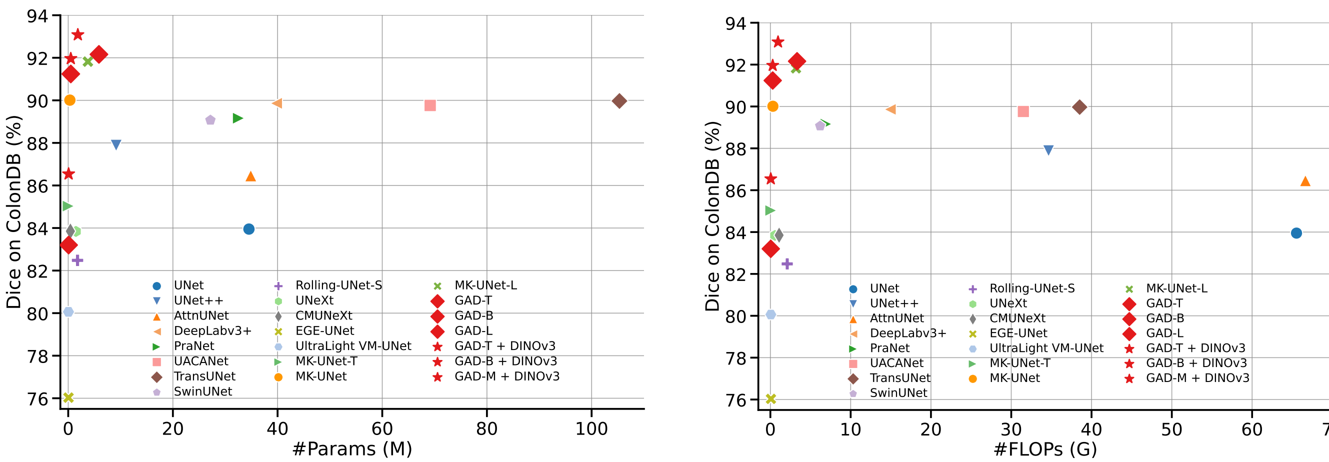

Fig. 1. Accuracy–efficiency comparison across PH^2, ISIC2018, CVC-ClinicDB, and CVC-ColonDB. The vertical axis reports the macro-average Dice score over the four datasets, while the horizontal axes show student-side parameters and FLOPs. Compared with existing lightweight segmentation networks, GAD-MambaUNet achieves a more favorable balance between segmentation accuracy and deployment cost. The DINOv3 teacher is used only during training and is excluded from deployment cost.

can still lack the global evidence needed when boundaries are incomplete, lesion regions are ambiguous, or distant anatomical structures share similar textures.

Visual state-space models provide a promising route to efficient global modeling. Mamba [10] models input-dependent sequences with linear complexity, and VMamba [11] extends selective scanning to two-dimensional visual features. VM-UNet [12] integrates visual state-space blocks into a U-shaped architecture for medical image segmentation. Compared with global self-attention, selective scanning offers a more efficient mechanism for long-range dependency modeling, making it attractive for lightweight segmentation networks.

To further reduce the cost of Vision Mamba, UltraLight VM-UNet [13] partitions deep features into multiple parallel channel groups and processes them using parallel Vision Mamba branches. This design reduces the parameter and computational cost of global modeling. However, partitioning features into groups also creates a representation-isolation issue. In conventional grouped multi-directional selective scanning, each direction–group branch is processed independently, and the responses are merged only after selective scanning. Consequently, complementary context from different traversal directions and channel groups cannot influence one another before fusion. This limitation motivates a more structured mechanism for direction–group interaction.

To address these issues, we propose *GAD-MambaUNet*, an asymmetric local–global U-shaped network for lightweight medical image segmentation. The architecture preserves multi-kernel convolution at high-resolution stages, where local texture extraction and boundary reconstruction are most important, and introduces grouped state-space blocks at deep low-resolution stages, where global reasoning is more economical. Within each state-space block, we propose Direction-Group Graph Selective Scan (DG-GSS), which represents each scan-direction and channel-group response as a graph node and enables structured information exchange among nodes sharing the same direction or the same channel-group identity before multi-directional fusion.

Beyond architecture design, compact segmentation models trained on limited medical datasets may still suffer from insufficient semantic abstraction. Inspired by the DINOv3-based gradient-adaptive distillation strategy in RT-DETRv4 [14], we adapt training-time foundation-model supervision to lightweight medical image segmentation. Specifically, a frozen DINOv3 teacher [15] provides dense semantic supervision for the aligned decoder feature, while Gradient-Adaptive Distillation (GAD) adjusts the distillation coefficient according to the gradient share of this decoder block. The teacher and projection head are discarded during inference, so the semantic supervision introduces no additional deployment cost.

To evaluate the proposed method, we conduct experiments on PH^2, ISIC2018, CVC-ClinicDB, and CVC-ColonDB. The main contributions of this work are summarized as follows:

- We develop an asymmetric local–global U-shaped architecture that preserves high-resolution multi-kernel convolution for boundary-sensitive local modeling and introduces grouped state-space modeling at deep semantic stages for efficient context aggregation.
- We propose DG-GSS, which explicitly exchanges information among direction–group responses through a factorized graph before multi-directional selective-scan fusion, alleviating the representation-isolation issue in grouped selective scanning.
- We adapt RT-DETRv4-style DINOv3 feature supervision and gradient-adaptive distillation to lightweight medical image segmentation, aligning teacher features with a deep decoder block while adding no teacher-side inference cost.
- We conduct extensive experiments on PH^2, ISIC2018,

CVC-ClinicDB, and CVC-ColonDB, showing that GAD-MambaUNet achieves a superior accuracy–efficiency balance over existing lightweight segmentation networks in terms of Dice score, parameters, and FLOPs, as illustrated in Fig. 1.

II. Related Work

A. Medical Segmentation and Lightweight U-Shaped Networks

U-Net [1] and its encoder–decoder variants remain a primary template for medical image segmentation because skip connections combine deep semantic features with spatially precise reconstruction. Task-specific methods strengthen this template through mechanisms such as reverse attention in PraNet [2] and uncertainty-aware context attention in UACANet [3]. TransUNet [4] instead introduces a Transformer encoder to enlarge the receptive field through global self-attention. These approaches establish the value of contextual reasoning, but their attention modules or pretrained encoders can be costly when the deployment budget is limited.

Lightweight U-shaped networks reduce this cost through compact channel schedules and efficient local operators. UNeXt [5] combines convolution with shifted MLP blocks, CMUNeXt [6] uses large kernels and skip fusion, EGE-UNet [7] develops group-enhanced attention, and EMCAD [8] employs efficient multi-scale convolutional decoding. MK-UNet [9] further uses parallel multi-kernel depth-wise convolutions, channel shuffle, and lightweight decoder attention to capture multi-scale boundary evidence. These lightweight CNN-based designs are effective for preserving local details, but they still mainly rely on local operators and may lack explicit long-range contextual modeling.

B. Visual State-Space Models

State-space models offer an efficient alternative to self-attention for long-range dependency modeling. Mamba [10] uses input-dependent state transitions with linear sequence complexity, while VMamba [11] extends selective scanning to visual feature maps through multiple two-dimensional traversal directions. VM-UNet [12] incorporates visual state-space blocks into a U-shaped segmentation architecture. For lightweight medical segmentation, UltraLight VM-UNet [13] further reduces Vision Mamba cost by partitioning deep features into parallel channel groups and processing them with parallel Vision Mamba branches.

In grouped multi-directional scans, direction–group branches are typically processed independently and merged only after selective scanning. Although this design reduces computation, it does not explicitly model the structural relationship between traversal directions and channel subspaces. This can limit the exchange of complementary directional cues and group-wise semantic responses before fusion.

C. Foundation-Model Distillation for Lightweight Segmentation

Strong Transformer encoders and vision foundation models provide rich semantic representations for medical image segmentation. Swin Transformer [16] and its medical segmentation variant Swin-Unet [17] model hierarchical visual features, while DINOv3 [15] provides dense features through large-scale self-supervised pretraining. However, directly deploying such large encoders conflicts with the parameter, memory, and latency budgets of lightweight systems.

Feature-level distillation offers a practical alternative: a frozen teacher provides dense supervision only during training, allowing a compact student to inherit semantic context without the teacher's deployment cost [18]. Most distillation settings use a fixed distillation coefficient. However, a fixed coefficient cannot adapt to the dynamic state of optimization: as training progresses, the gradient contribution of the teacher-supervised deep student feature to the overall network can change, so the same coefficient may yield insufficient teacher supervision or impose an overly strong constraint on the primary task objective.

To address this dynamic balancing problem, RT-DETRv4 [14] proposes a gradient-aware distillation strategy for real-time detection. In its reported configuration, a frozen DINOv3-ViT-B teacher supervises deep detector features through a Deep Semantic Injector, and Gradient-guided Adaptive Modulation adjusts the semantic-transfer strength using gradient norm ratios. Inspired by this strategy, our work adapts DINOv3-based gradient-adaptive distillation to lightweight medical image segmentation, where teacher features are aligned with a deep decoder representation and discarded during inference.

III. Methodology

We describe the proposed GAD-MambaUNet by first introducing its core building block and then integrating it into an asymmetric U-shaped segmentation architecture. As shown in Fig. 2, the left part presents the overall student network and the training-time DINOv3 teacher, while the right part details the proposed block-level design and the internal structure of Direction-Group Graph Selective Scan (DG-GSS). We first present DG-GSS, which enables structured interaction among scan directions and channel groups. Then, we describe how DG-GSS is embedded into the overall GAD-MambaUNet architecture. Finally, we introduce the training-time DINOv3 supervision with Gradient-Adaptive Distillation (GAD), where the teacher model is used only during training and removed during inference.

A. Direction-Group Graph Selective Scan

As illustrated in the lower-right part of Fig. 2, DG-GSS is designed to enhance grouped multi-directional selective scanning by introducing graph-based interaction among direction–group responses. Grouped multi-directional selective scanning reduces the cost of Vision Mamba by dividing feature channels into multiple groups and processing them along several

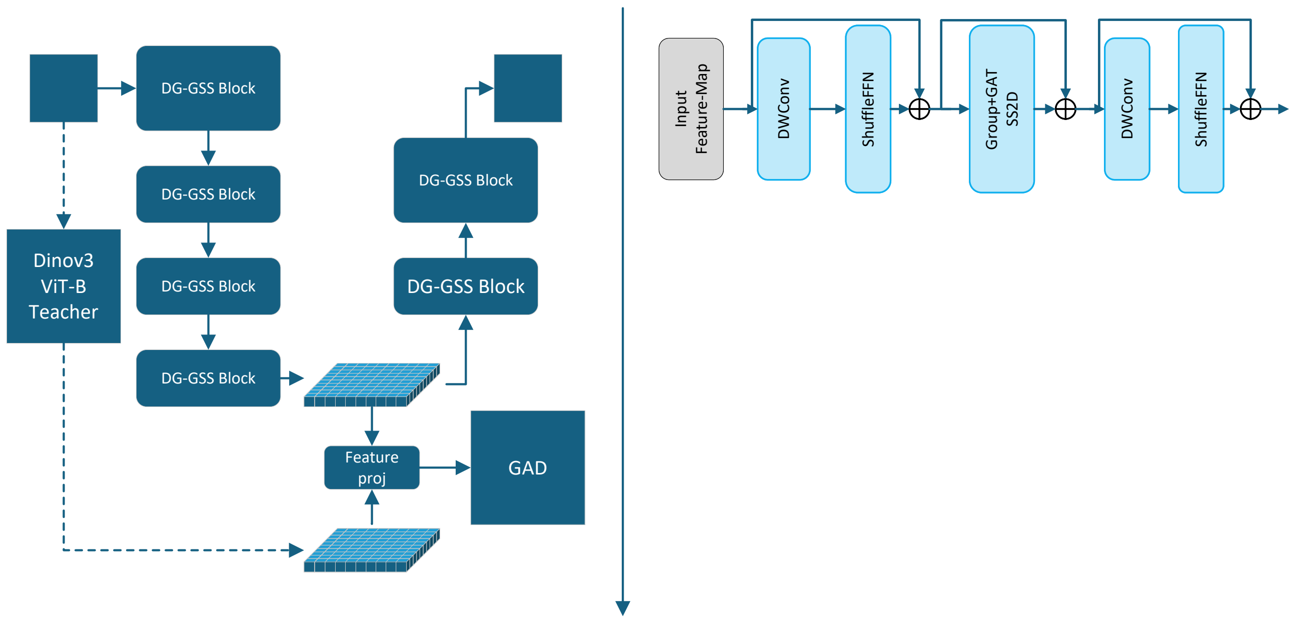

Fig. 2. Overall architecture of GAD-MambaUNet. High-resolution stages retain multi-kernel local modeling, whereas deep stages use grouped state-space blocks with DG-GSS. During training, a frozen DINOv3 teacher supervises the first decoder block and GAD regulates the distillation coefficient according to its gradient share. The teacher and projection head are removed at inference.

traversal directions. However, in conventional grouped scanning, each direction–group branch is processed independently and fused only after selective scanning. This independent processing limits the exchange of complementary information across scan directions and channel subspaces. To address this issue, DG-GSS regards each scan-direction and channel-group response as a graph node and performs structured message passing before the final multi-directional fusion.

Let $\mathbf{X} \in \mathbb{R}^{B\times D_i\times h\times w}$ denote the inner state-space feature of DG-GSS, where D_i is the inner channel dimension. We adopt four Cross2D traversal directions and partition the inner channels into M groups. With $d = D_i/M$ and $L = hw$, cross scanning produces

$$\mathbf{X}_{\text{scan}} \in \mathbb{R}^{B\times K\times M\times d\times L}, \qquad K = 4. \tag{1}$$

For each direction–group pair, the selective-scan parameters $(\Delta, \mathbf{B}, \mathbf{C})$ are generated by independent grouped projections, and selective scanning is applied along the corresponding sequence. The directional outputs are then realigned to the original spatial coordinate system:

$$\mathbf{Y} \in \mathbb{R}^{B\times K\times M\times d\times h\times w}. \tag{2}$$

Each tensor $\mathbf{Y}_{k,m}$ corresponds to the response of scan direction k and channel group m. We regard every direction–group response as a graph node and obtain its node descriptor by global average pooling:

$$\mathbf{r}_{k,m} = \frac{1}{hw}\sum_{u=1}^{h}\sum_{v=1}^{w}\mathbf{Y}_{k,m,:,u,v}. \tag{3}$$

After layer normalization and linear projection, the node descriptor is transformed into $\mathbf{h}_{k,m}$.

In our implementation, we adopt a factorized direction-group graph, where two nodes are connected if they share the same scan direction or the same channel group, and self-connections are included. For two connected nodes i and j, additive graph-attention logits [19] are computed as

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}_s^{\top}\mathbf{h}_i + \mathbf{a}_t^{\top}\mathbf{h}_j\right). \tag{4}$$

The attention coefficients are obtained by masked softmax over the neighborhood:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{j'\in\mathcal{N}(i)}\exp(e_{ij'})}, \qquad j \in \mathcal{N}(i). \tag{5}$$

Graph-based message passing is then performed on the full spatial feature maps:

$$\widetilde{\mathbf{Y}}_i = \mathbf{Y}_i + \gamma\sum_{j\in\mathcal{N}(i)}\alpha_{ij}V(\mathbf{Y}_j), \tag{6}$$

where $V(\cdot)$ is a shared 1×1 value projection and γ is a learnable scaling parameter initialized to zero. This initialization makes DG-GSS start from the original grouped selective-scan behavior and progressively learn direction–group information exchange during training.

Finally, the enhanced group responses are concatenated along the channel dimension for each direction, and the four aligned directional feature maps are summed:

$$\mathbf{Z} = \sum_{k=1}^{K} \mathrm{Concat}_{m=1}^{M} \left(\widetilde{\mathbf{Y}}_{k,m} \right). \tag{7}$$

Output normalization is then applied to obtain the DG-GSS output. In this way, DG-GSS preserves the efficiency of grouped multi-directional selective scanning while explicitly modeling the structural relationship between scan directions and channel groups.

B. GAD-MambaUNet Architecture

After defining DG-GSS, we embed it into a lightweight block and integrate the block into an asymmetric U-shaped segmentation network. As shown in the upper-right part of Fig. 2, the DG-GSS block consists of three residual components: a pre-scan local mixing stage, the proposed DG-GSS layer, and a post-scan local refinement stage. Given an input feature $\mathbf{x}$, the block is formulated as

$$\begin{aligned} \mathbf{u}_1 &= \mathbf{x} + \mathrm{FFN}_0(\mathrm{DW}_0(\mathbf{x})), \\ \mathbf{u}_2 &= \mathbf{u}_1 + \text{DG-GSS}(\mathbf{u}_1), \\ \mathbf{y} &= \mathbf{u}_2 + \mathrm{FFN}_1(\mathrm{DW}_1(\mathbf{u}_2)), \end{aligned} \tag{8}$$

where $\mathrm{DW}(\cdot)$ denotes depth-wise convolution for local mixing and $\mathrm{FFN}(\cdot)$ denotes a lightweight feed-forward refinement module. The local operators enhance short-range texture and channel interactions, while DG-GSS provides efficient long-range context aggregation through direction–group graph interaction.

The overall GAD-MambaUNet follows a five-level encoder–decoder architecture, as shown in the left part of Fig. 2. Given an input image $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$, the network predicts a binary segmentation logit map $\mathbf{z} \in \mathbb{R}^{1\times H\times W}$. In the base configuration, the channel widths are set to $[16, 32, 64, 96, 160]$. To balance local boundary preservation and global context modeling, the first three encoder stages and the last three decoder stages use lightweight multi-kernel inverted residual (MKIR) blocks inherited from MK-UNet [9], while the two deepest encoder stages and the first two decoder stages use DG-GSS blocks. Resolution and channel changes are performed outside these blocks through depth-wise downsampling with point-wise projection or point-wise projection followed by bilinear upsampling, keeping the feature width fixed inside each block.

For decoder reconstruction, skip connections transfer high-resolution encoder features to the corresponding decoder stages. Before fusion, a grouped attention gate filters the skip feature according to the decoder context, reducing irrelevant background responses. Let $\mathbf{g}$ denote the upsampled decoder feature and $\mathbf{s}$ the encoder feature at the same resolution. The gated skip feature is computed as

$$\boldsymbol{\alpha} = \sigma!\left(\psi!\left(\phi(W_g\mathbf{g} + W_s\mathbf{s})\right)\right), \qquad \hat{\mathbf{s}} = \boldsymbol{\alpha} \odot \mathbf{s}, \tag{9}$$

where W_g and W_s are grouped convolutions, ϕ is ReLU, and ψ maps the joint response to a spatial gate. The filtered skip feature $\hat{\mathbf{s}}$ is then fused with the decoder feature. The first decoder block operates at the deepest decoder level and provides the aligned student feature for the training-time DINOv3 supervision described in the next subsection.

C. Training-Time DINOv3 Supervision with GAD

Although DG-GSS improves contextual modeling inside the compact student network, training a lightweight segmentation model on limited medical datasets may still lead to insufficient semantic abstraction. Inspired by the DINOv3-based gradient-adaptive distillation strategy in RT-DETRv4 [14], we adapt training-time foundation-model supervision to lightweight medical image segmentation. A frozen DINOv3 ViT-B/16 model [15] is used as the semantic teacher only during training, and the teacher branch is removed during inference.

As described in the previous subsection, the student feature used for distillation is extracted after the first decoder block and before the first upsampling operation. This feature is semantically rich, spatially compact, and computationally inexpensive for feature alignment. Let $\mathbf{F}_s$ denote the projected student feature, and let $\mathbf{F}_t$ denote the corresponding DINOv3 teacher feature. The student feature is projected to the teacher dimension by a 1×1 convolution followed by batch normalization when their channel dimensions are different. If necessary, the teacher feature is bilinearly resized to match the spatial resolution of $\mathbf{F}_s$.

After spatial alignment, both feature maps are flattened into N spatial tokens and ℓ_2-normalized along the channel dimension. We use a cosine feature distillation loss:

$$\mathcal{L}_{\mathrm{dist}} = \frac{1}{N} \sum_{n=1}^{N} \left[1 - \frac{\mathbf{f}_{s,n}^{\top} \mathbf{f}_{t,n}}{\|\mathbf{f}_{s,n}\|_2 \|\mathbf{f}_{t,n}\|_2 + \epsilon} \right], \tag{10}$$

where $\mathbf{f}_{s,n}$ and $\mathbf{f}_{t,n}$ denote the n-th student and teacher tokens, respectively, and ϵ is a small constant for numerical stability.

The segmentation objective follows the structure-aware weighted BCE and weighted IoU loss [2]:

$$\mathcal{L}_{\mathrm{seg}} = \mathcal{L}_{\mathrm{wbce}} + \mathcal{L}_{\mathrm{wiou}}. \tag{11}$$

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{seg}} + \lambda_e \mathcal{L}_{\mathrm{dist}}, \tag{12}$$

where λ_e is the distillation coefficient at epoch e.

Using a fixed distillation coefficient assumes that the proper teacher supervision strength remains unchanged throughout training. However, the optimization state of the student changes over time, and the gradient contribution of the teacher-supervised decoder block may become either too weak or too dominant. To balance semantic supervision and the primary segmentation objective, GAD dynamically updates λ_e according to the gradient share of the aligned decoder block. After back-propagation and before gradient clipping, we compute

$$q_e = 100 \times \frac{\sum_{\theta_i \in \Theta_a} \|\nabla_{\theta_i} \mathcal{L}\|_1}{\sum_{\theta_j \in \Theta} \|\nabla_{\theta_j} \mathcal{L}\|_1 + \epsilon}, \tag{13}$$

where Θ_a denotes the parameters of the aligned decoder block and Θ denotes all trainable student parameters.

Given a target gradient-share interval $[\rho - \delta, \rho + \delta]$, GAD adjusts the next-epoch distillation coefficient only when q_e falls outside this interval. Let q^* be the nearest target boundary, and define $p_e = q_e/100$ and $p^* = q^*/100$. The odds-ratio adjustment is computed as

$$r_e = \frac{p^*(1 - p_e)}{p_e(1 - p^*) + \epsilon}. \tag{14}$$

The distillation coefficient for the next epoch is updated by

$$\lambda_{e+1} = \mathrm{clip}\left(\lambda_e \, \mathrm{clip}(r_e, r_{\min}, r_{\max}), \lambda_{\min}, \lambda_{\max}\right), \tag{15}$$

where $r_{\min}$ and $r_{\max}$ bound the per-epoch adjustment, and $\lambda_{\min}$ and $\lambda_{\max}$ constrain the overall distillation strength.

Training begins with a segmentation-only burn-in stage, followed by a linear distillation warm-up. GAD is activated after warm-up to adaptively regulate the teacher contribution. During inference, the DINOv3 teacher, the feature projection head, and the GAD update are all discarded. Therefore, the proposed training-time supervision improves the compact student without increasing deployment cost.

IV. Experiments and Results

A. Datasets and Metrics

We evaluate GAD-MambaUNet on four public binary medical image segmentation datasets covering two representative segmentation tasks. For skin lesion segmentation, we use PH^2 [20], which contains 200 dermoscopic images with expert lesion annotations, and ISIC2018 [21], which contains 2,594 dermoscopic images with pixel-level lesion masks. For polyp segmentation, we use CVC-ClinicDB [22], which contains 612 colonoscopic polyp images, and CVC-ColonDB [23], [24], which contains 379 colonoscopic images collected from different video sequences. These datasets include diverse imaging conditions, lesion appearances, object scales, boundary ambiguities, and foreground–background contrasts, providing a comprehensive evaluation of the robustness of lightweight segmentation models.

We split each dataset into training, validation, and test subsets with a ratio of 80:10:10. The validation set is used for model selection, and the test set is used for final performance reporting. Dice score is used as the primary segmentation metric, and Intersection over Union (IoU) is also reported when applicable. To evaluate model efficiency, we report the number of trainable parameters and floating-point operations (FLOPs). Since the DINOv3 teacher is used only during training and is discarded during deployment, it does not introduce additional inference computation. Therefore, all efficiency metrics are computed using the student network alone.

B. Implementation Protocol

All experiments are implemented in PyTorch and conducted under the same training and evaluation settings for fair comparison. Images from PH^2 and ISIC2018 are resized to 256×256, while images from CVC-ClinicDB and CVC-ColonDB are resized to 352×352. The validation set is used for checkpoint selection, and the corresponding test performance is reported.

For GAD-MambaUNet, we use AdamW as the optimizer with an initial learning rate of 5×10^{-4}, a weight decay of 10^{-4}, and cosine learning-rate decay to 10^{-6}. All models are trained for 400 epochs with a batch size of 8, gradient clipping at 0.5, and standard data augmentation. The segmentation objective is the sum of weighted binary cross-entropy and weighted IoU losses. Unless otherwise specified, each configuration is repeated over five independent runs, and the mean test performance is reported.

For training-time semantic supervision, we use frozen DINOv3 ViT-T/16 and ViT-B/16 models as semantic teachers in different experimental settings. The student feature after the first decoder block is aligned to the teacher feature through the projection head described in Section **??**. DINOv3 distillation starts at epoch 10 and is linearly warmed up for 20 epochs. GAD is activated at epoch 30 and adaptively controls the distillation coefficient until epoch 360. During inference, the teacher branch, projection head, and GAD update are discarded, so the reported parameters and FLOPs are computed using only the student network.

C. Reference Baseline Analysis

Table I summarizes the reference landscape together with the completed CVC-ClinicDB and CVC-ColonDB scaling runs of GAD-MambaUNet. Fig. 1 further visualizes the average Dice–efficiency trade-off against student-side parameters and FLOPs. Heavy encoder-decoder and Transformer-based models, such as TransUNet, provide strong Dice scores but require substantially more parameters and computation. Lightweight CNN and state-space models reduce the computational footprint, but their performance varies across modalities: some are effective on skin lesions but degrade on colonoscopic polyp datasets where boundary ambiguity and background similarity are more severe.

MK-UNet provides a strong local-convolutional baseline with a compact parameter count and competitive Dice scores across the four datasets. This makes it an appropriate reference point for our design. Without DINOv3 supervision, performance rises sharply from Tiny to Base and then largely saturates, with Large achieving the highest validation-selected ColonDB test Dice. Adding DINOv3 supervision improves the Medium variant to the best reported ColonDB score, while retaining the same student-side inference cost. The remaining experiments will determine whether GAD further improves this trade-off.

V. Ablation Study

The ablation study is designed to separate the contribution of the three main design choices: deep grouped state-space modeling, direction-group graph interaction, and gradient-adaptive DINOv3 distillation. Table II lists the planned variants. All variants will use the same data split, preprocessing,

TABLE I
COMPARISON ON FOUR DATASETS. NON-PH^2 BASELINE VALUES ARE TAKEN FROM THE ORIGINAL MK-UNET PAPER UNDER ITS REPORTED PROTOCOL. DICE SCORES ARE REPORTED IN %. PH^2, CLINICDB, AND COLONDB ENTRIES FOR MK-UNET AND GAD-MAMBAUNET ARE MEAN TEST DICE AT THE VALIDATION-SELECTED CHECKPOINT OVER REPEATED RUNS. DINOV3 IS USED ONLY DURING TRAINING AND ADDS NO INFERENCE COST.

Method	Params(M)	FLOPs(G)	Throughput(/s)	PH^2	ISIC2018	ClinicDB	ColonDB	Avg.
U-Net	34.53	65.53	119.03	–	86.67	91.43	83.95	–
PraNet	32.55	6.93	64.10	–	88.46	91.71	89.16	–
UACANet	69.16	31.51	43.28	–	88.72	93.29	89.76	–
TransUNet	105.32	38.52	65.35	–	89.04	93.18	89.97	–
UNeXt	1.47	0.57	172.41	–	87.78	90.20	83.84	–
CMUNeXt	0.418	1.09	175.43	–	87.51	92.82	83.85	–
EGE-UNet	0.054	0.072	101.01	–	86.95	84.76	76.03	–
UltraLight VM-UNet	0.050	0.060	98.03	–	87.85	87.11	80.06	–
MK-UNet-T	0.027	0.062	140.85	94.56	88.19	91.26	85.03	–
MK-UNet	0.316	0.314	138.89	94.39	88.74	93.48	90.01	–
MK-UNet-L	3.76	3.19	122.62	94.56	89.25	93.85	91.82	–
GAD-MambaUNet-Tiny (Ours)	0.076	0.051	–	94.43	88.27	92.47	83.20	–
GAD-MambaUNet-Small (Ours)	0.199	0.117	–	94.40	88.63	93.85	90.17	–
GAD-MambaUNet-Base (Ours)	0.493	0.291	–	**94.90**	88.91	94.04	91.24	–
GAD-MambaUNet-Medium (Ours)	1.839	0.955	–	94.71	89.13	94.07	91.35	–
GAD-MambaUNet-Large (Ours)	5.871	3.325	–	94.88	89.34	**94.70**	**92.16**	–
GAD-MambaUNet-Tiny + DINOv3 (Ours)	0.076	0.051	–	94.61	88.61	93.09	86.54	–
GAD-MambaUNet-Small + DINOv3 (Ours)	0.199	0.117	–	94.73	88.94	94.16	91.51	–
GAD-MambaUNet-Base + DINOv3 (Ours)	0.493	0.291	–	95.16	89.06	94.17	91.96	–
GAD-MambaUNet-Medium + DINOv3 (Ours)	1.839	0.955	–	95.21	89.11	94.52	**93.08**	–
GAD-MambaUNet-Large + DINOv3 (Ours)	5.871	3.325	–	95.37	89.53	95.08	93.01	–

TABLE II
PLANNED ABLATION VARIANTS FOR GAD-MAMBAUNET. VALUES ARE LEFT BLANK UNTIL THE CORRESPONDING EXPERIMENTS ARE COMPLETED.

Variant	BUSI	ISIC2018	ClinicDB	ColonDB	Avg.
Local student baseline	–	–	–	–	–
+ Grouped Mamba blocks	–	–	–	–	–
+ DG-GSS	–	–	–	–	–
+ DINOv3 fixed distillation	–	–	–	–	–
+ GAD distillation	–	–	–	–	–

segmentation loss, and training schedule, so the measured differences can be attributed to architectural and distillation changes.

A. Effect of Direction-Group Global Modeling

The first ablation compares the local student baseline with variants that insert grouped Mamba blocks and DG-GSS into the deep encoder-decoder stages. This comparison tests whether global context is beneficial when introduced only at low spatial resolutions. A positive result would indicate that long-range dependencies can be added without disturbing the high-resolution local features used for boundary reconstruction.

B. Effect of DINOv3 Distillation

The second ablation compares training without distillation, training with a fixed DINOv3 distillation coefficient, and training with GAD. Fixed distillation measures whether dense foundation-model features provide useful semantic regularization. GAD further measures whether adapting the teacher weight according to the student's gradient distribution stabilizes the transfer and improves the final mask quality.

VI. CONCLUSION

This paper presents GAD-MambaUNet, a lightweight medical image segmentation framework that combines high-resolution multi-kernel local modeling, deep direction-group state-space interaction, and gradient-adaptive DINOv3 feature distillation. The design keeps inference compact by using the foundation model only during training, while targeting the common failure modes of binary medical segmentation: weak boundaries, appearance ambiguity, and insufficient global context. Initial CVC-ClinicDB and CVC-ColonDB scaling results without DINOv3 show that strong accuracy is attainable with sub-megaparameter variants, while Large gives the strongest reported validation-selected ColonDB result. The remaining multi-dataset and distillation ablations will complete the evaluation.

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